

SABHA, Libyan Arab Jamahiriya

WMO: 621240

Lat: 26.987N

Lon: 14.473E

Elev: 1427

StdP: 13.95

Time Zone: 2.00 (E02)

Period: 96-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.7	42.6	12.3	10.9	69.8	16.5	13.3	68.6	37.7	63.5	31.6	65.6	4.3	100	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	24.8	108.5	68.2	106.2	68.1	104.3	67.6	72.7	100.5	71.4	99.6	70.2	98.4	12.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	90.9	87.9	61.0	84.4	87.7	59.0	78.7	87.8	37.2	100.6	36.0	99.4	34.9	98.4	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 30.3	(b) 25.8	(c) 22.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 30.3	(b) 25.8	(c) 22.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.4	56.5	59.1	67.6	77.4	85.1	89.9	89.4	89.1	87.8	77.9	66.6	57.6
	DBStd	13.94	5.13	6.63	7.86	6.99	6.98	5.25	3.86	3.14	4.70	6.22	5.53	5.78
	HDD50	7	3	1	0	0	0	0	0	0	0	0	0	3
	HDD65	823	271	194	68	4	0	0	0	0	0	0	43	243
	CDD50	9279	204	255	544	822	1089	1196	1222	1212	1133	866	497	239
	CDD65	4622	8	28	147	376	624	746	757	747	683	402	90	14
	CDH74	67308	74	326	1696	4628	9222	11929	12055	11717	10127	4530	850	154
	CDH80	41344	8	117	711	2432	5788	8013	7875	7524	6355	2232	264	25
(14) Wind	WSAvg	11.3	9.8	9.6	11.8	13.6	14.0	13.0	12.3	11.5	11.5	10.7	9.0	8.3
(15) Precipitation	PrecAvg	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecMax	2.0	1.0	0.6	0.6	0.3	0.3	0.6	0.0	0.0	0.2	0.4	0.4	0.7
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.4	0.2	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.1	0.1	0.1	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.6	89.6	96.7	104.1	109.5	113.3	109.8	108.3	108.3	103.4	91.7	83.1
		MCWB	56.3	60.0	61.8	65.9	67.2	70.1	68.0	68.6	68.7	66.2	61.4	55.4
	2%	DB	75.2	83.3	91.7	99.5	105.7	109.1	106.8	105.3	105.7	97.7	87.2	78.8
		MCWB	52.7	57.8	61.3	63.8	67.0	68.5	68.8	68.5	68.8	65.5	60.1	53.9
	5%	DB	71.8	76.9	87.6	96.2	103.1	106.2	104.2	102.8	103.4	94.0	82.7	73.7
		MCWB	52.3	54.7	58.7	63.3	66.3	67.6	68.7	67.9	67.6	64.4	58.5	53.4
	10%	DB	68.1	71.8	83.6	92.5	100.1	103.9	101.7	100.7	100.8	91.0	78.9	69.9
		MCWB	51.1	51.6	56.9	61.4	65.1	67.0	67.8	67.5	67.4	63.5	57.2	52.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.0	61.5	64.6	69.9	72.0	73.6	74.6	74.0	73.9	70.1	66.7	59.6
		MCDB	75.5	87.3	91.7	99.6	102.5	106.6	100.8	98.6	99.2	94.0	82.6	76.9
	2%	WB	55.9	58.2	62.1	66.8	69.9	71.5	72.5	71.7	71.7	68.1	62.9	56.7
		MCDB	72.1	80.1	88.2	94.1	100.0	104.0	100.1	98.6	97.8	91.0	83.1	72.5
	5%	WB	53.8	55.6	59.8	64.6	68.1	69.9	71.0	70.3	70.2	66.4	60.4	54.9
		MCDB	68.5	74.3	85.2	90.4	97.1	102.4	98.6	97.3	97.3	88.7	78.4	70.2
	10%	WB	51.9	53.2	57.8	62.9	66.4	68.3	69.6	69.3	68.7	64.9	58.5	53.2
		MCDB	65.8	69.7	80.9	87.9	94.7	99.8	97.2	96.4	96.4	87.0	75.4	68.2
(35) Mean Daily Temperature Range	5% DB	MDBR	22.3	23.3	25.4	25.0	24.5	24.8	23.2	22.9	23.7	22.9	23.1	23.1
		MCDBR	26.4	28.4	30.2	28.9	26.4	26.8	25.4	24.9	25.9	25.4	26.2	28.0
		MCWBR	13.0	13.3	13.8	11.9	10.2	10.4	9.8	9.1	9.6	10.1	11.6	13.5
		MCDBR	24.1	26.5	28.0	27.1	25.5	26.2	23.2	22.8	24.2	23.3	23.6	24.7
	5% WB	MCWBR	13.6	13.4	13.6	12.3	11.3	11.5	10.5	10.1	11.0	10.2	12.4	13.6
(40) Clear-Sky Solar Irradiance	taub		0.370	0.409	0.497	0.611	0.645	0.587	0.489	0.496	0.536	0.486	0.417	0.374
	taud		2.084	1.970	1.746	1.549	1.507	1.628	1.851	1.849	1.751	1.846	1.976	2.087
	Ebn at Noon		272	271	253	229	221	233	257	255	239	243	253	264
	Edh at Noon		44	54	72	89	94	82	66	65	70	60	49	43
(44) All-Sky Solar Radiation	RadAvg		1365	1637	1992	2192	2333	2504	2540	2363	2097	1710	1419	1253
	RadStd		69	93	86	84	96	63	24	30	50	67	39	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page